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44	biomedcentral.com	<1	Internet Data
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47	From face-to-face to remote teaching during COVID-19 Lecturers at private coll By Ashika Maharaj, Percyval Baya, Yr-2025,6,5	<1	Publication
48	ijeeedu.org	<1	Internet Data
49	journal.esrgroups.org	<1	Publication
50	moam.info	<1	Internet Data
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54	www.informatica.si	<1	Publication

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A PROJECT REPORT ON

**MindTrace: Real-Time Emotion Recognition
and Engagement Tracking System**

SUBMITTED TO THE SAVITRIBAI PHULE PUNE UNIVERSITY, PUNE

FOR THE PARTIAL FULFILLMENT OF THE REQUIREMENTS

FOR THE AWARD OF THE DEGREE

BACHELOR OF ENGINEERING

(Computer Engineering)

BY

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ABSTRACT

Real-time emotion recognition and engagement tracking in educational and professional environments remains a significant challenge, as traditional assessment methods are subjective and impractical at scale. MindTrace addresses this by providing a fully automated system that operates through a standard webcam without any additional hardware. It uses MediaPipe Face Mesh for face detection, feeding regions into a ResNet-18 model fine-tuned on the RAF-DB dataset to classify seven emotional states: Angry, Disgust, Fear, Happy, Neutral, Sad, and Surprise. A composite engagement score (0–100) is computed by combining the detected emotion, head pose estimation, and blink-rate analysis.

The backend is built on FastAPI following a stateless, client-driven architecture where individual webcam frames are sent via REST endpoints for inference, enabling seamless cloud deployment on Hugging Face Spaces without dependency on local hardware. All session data, emotion logs, and engagement metrics are persisted in a MongoDB database to support historical analytics and reporting. The React frontend delivers a professional analytics dashboard featuring live emotion overlays, session timelines, emotion distribution charts, and a multi-user admin panel. Security is handled through JWT-based authentication and bcrypt password hashing with role-based access control for students and administrators. The ResNet-18 model achieves over 85% validation accuracy on the RAF-DB benchmark, and the overall system demonstrates low-latency real-time performance suitable for deployment in e-learning platforms, corporate training environments, and remote work settings.

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KEYWORDS

Emotion Recognition, Facial Action Units, ResNet-18, RAF-DB Dataset, MediaPipe Face Mesh, Head Pose Estimation, Engagement Tracking, FastAPI, MongoDB, React, JWT Authentication, Deep Learning, Transfer Learning, Real-Time Processing, Student Engagement

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CHAPTER 1
INTRODUCTION
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1.1 Motivation

The rapid shift toward remote learning, online education, and digital workplaces has created an urgent need for tools that can objectively measure learner and employee engagement. In a traditional classroom or meeting room, an instructor or manager can intuitively observe whether participants are attentive, confused, frustrated, or simply disengaged. However, in virtual environments — or even large physical classrooms — such qualitative assessment becomes impractical at scale.

Research consistently shows that emotional state is one of the strongest predictors of learning outcomes and work productivity. A student experiencing confusion or frustration is likely to disengage if not promptly supported, while a student expressing curiosity or happiness is more likely to retain information. Despite this, most e-learning and remote work platforms today provide no mechanism for real-time affective feedback.

Advances in deep learning, particularly in facial action recognition and transfer learning, now make it computationally feasible to detect human emotions in real time using only a standard webcam. MindTrace leverages these advances to build a non-invasive, automated engagement tracking system that works entirely through the user's existing camera, requiring no wearable sensors, specialised hardware, or manual input.

1.2 Problem Definition and Objectives

1.2.1 Problem Definition

To design and implement a real-time emotion recognition and engagement tracking system using deep learning that analyses a user's facial expressions through a webcam, computes a continuous focus/engagement score, persists session-level analytics in a cloud database, and presents insights through an interactive web dashboard — all without requiring any specialised hardware beyond a standard computer camera.

1.2.2 Objectives

- To detect and classify the emotional state of a user in real time from webcam frames into seven categories: Angry, Disgust, Fear, Happy, Neutral, Sad, and Surprise.
- To estimate head pose (pitch, yaw, roll) and detect attentional distraction when a user looks away from the screen.
- To compute a composite engagement/focus score (0–100) by combining emotional

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state, head pose, and blink rate signals.

- To persist all session data, emotion logs, and engagement metrics in a MongoDB database for historical analysis and reporting.
- To provide a React-based analytics dashboard offering live emotion overlays, session timelines, emotion distribution charts, and an admin panel for multi-user monitoring.
- To secure the system with JWT-based user authentication and bcrypt password hashing, supporting role-based access control for students and administrators.
- To deploy the backend on a cloud platform (Hugging Face Spaces) and the frontend on Vercel, enabling fully remote, hardware-independent operation.

1.3 Project Scope

MindTrace is designed for deployment in the following contexts:

- Educational Institutions — Monitoring student engagement during online lectures, self-paced video courses, or e-assessment sessions to identify students at risk of disengagement.
- Corporate Training — Tracking employee attention and emotional response during virtual onboarding, compliance training, or skill-development workshops.
- Remote Work Environments — Providing managers with aggregated, anonymised team engagement metrics during video conferences or collaborative sessions.
- Research — Serving as a data collection and annotation platform for affective computing researchers studying the relationship between emotion and cognitive load.

The system operates entirely client-side for frame capture and relies on a RESTful FastAPI backend for model inference, keeping user data private and processing load distributed. The modular architecture allows individual components — the emotion model, engagement algorithm, or frontend dashboard — to be upgraded independently as the project evolves.

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CHAPTER 2
LITERATURE SURVEY

4

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9 This chapter reviews existing research and literature in the fields of deep learning-based facial expression recognition and human fatigue detection. A summary of the most relevant studies is presented in the tables below, followed by a brief analysis of how these works informed the development of MindTrace.

Table 2.1: Literature Survey of Facial Emotion and Fatigue Detection (Part 1)

No.	Title & Source	Techniques	Key Contribution	Pros	Cons
1	A Survey of Deep Learning FER Research – IEEE Affective Computing, 2023	CNN, ResNet, Attention	Comprehensive review of FER models (FER-2013, AffectNet). Discusses illumination and class imbalance.	Wide architecture coverage.	Lacks real-time integration.
2	Improved FER Model Based on a Novel Deep Network – Elsevier, 2024	CNN, Batch Norm, Skip Conn.	Lightweight CNN with opti-		

mized residual
links for high
accuracy on
RAF-DB.

Low la-
tency;
embedded
ready.

Fails
in poor
lighting/
occlusion.

3 Advancements
in Driver Fa-
tigue Detection
– Springer, 2024

EAR, Facial
Landmark
Tracking
Hybrid system
combining
landmarks
and blink
pattern for
early fatigue
estimation.

Effective
for safety
systems.

Restricted
to driver
domain.

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Table 2.2: Literature Survey of Facial Emotion and Fatigue Detection (Part 2)

No. Title &

Source

Techniques Key Contri-
bution

Pros Cons

4 Non-Invasive

Fatigue Detec-
tion via Facial
Parameters

– ACM HCI,

2025

Emotion

Tracking,

Parameter

Ext.

Fatigue detec-
tion using fa-
cial parameter
variations for
adaptive inter-
action.

User-
friendly;

non-
contact.

Needs sta-
ble face po-
sition.

5 Fatigue Mon-
itoring Using
Wearables and

AI – IEEE

Sensors, 2025

CNN + Sen-
sor Fusion

Multimodal

framework

combining

sensors (heart
rate, blink)

with CNN

emotion recog-

dition.

High precision and robustness.

Relies on external sensors.

2.1 Analysis of Existing Literature

The reviewed literature highlights a significant transition in the field of affective computing:

- **From Pure Classification to Real-Time Systems:** Early works focused primarily on achieving high accuracy on static datasets. Recent trends, as seen in the work from Elsevier (2024), prioritize lightweight architectures suitable for deployment.
- **Introduction of Fatigue Metrics:** The integration of biometric signals like Eye Aspect Ratio (EAR), documented by Springer (2024), provides a deterministic way to measure focus and fatigue that complements emotional state analysis.
- **Multimodal vs. Visual-Only:** While multimodal systems (IEEE Sensors, 2025) offer high precision, the hardware complexity of wearables limits their adoption in standard e-learning environments. This justifies MindTrace's focus on purely visual, non-invasive monitoring.

MindTrace builds upon these foundations by combining the lightweight residual learning approach for emotion classification with the non-invasive facial parameter extraction

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for engagement scoring, delivering a unified real-time system that operates on standard consumer hardware.

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CHAPTER 3
SOFTWARE REQUIREMENTS SPECIFICATION
8

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3.1 Overall Description

3.1.1 Product Perspective

MindTrace is a standalone web-based application that integrates a deep learning inference backend with a browser-based frontend. It does not depend on third-party emotion analysis APIs; all inference is performed by the custom-trained ResNet-18 model hosted on the cloud. The system interacts with a MongoDB database for data persistence and with the client's webcam for real-time frame capture.

3.1.2 Product Functions

- Real-time facial emotion classification into seven categories.
- Head pose estimation to detect attention direction.
- Blink-rate measurement for fatigue detection.
- Composite engagement score computation (0–100 scale).
- Session-wise logging of emotion and engagement data.
- Interactive analytics dashboard with charts and timelines.
- Role-based user management (student and administrator roles).
- Secure user authentication with JWT and bcrypt.

3.1.3 User Classes and Characteristics

- **Students / Employees:** Use the system during sessions to have their engagement monitored. **45** They can view their own session history and emotion analytics through the dashboard.
- **Administrators / Instructors:** Have elevated access to view aggregated engagement reports for all users under their purview and manage user accounts.

3.1.4 Operating Environment

The system operates in the following environment:

- **Backend:** Python 3.10+, FastAPI, deployed on Hugging Face Spaces (Linux container).
- **Database:** MongoDB Atlas (cloud-hosted).
- **Frontend:** React + Vite, deployed on Vercel.

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- **Client:** Any modern web browser (Chrome, Firefox, Edge) with webcam access.

3.2 Functional Requirements

3.2.1 User Authentication

- The system shall allow users to register with a username, email, password, and role.
- Passwords shall be hashed using bcrypt before storage.
- On login, the system shall issue a signed JWT token valid for a configurable duration.
- Protected endpoints shall reject requests with invalid or expired tokens with an HTTP 401 response.

3.2.2 Emotion Detection

- The system shall accept a Base64-encoded webcam frame via a REST endpoint.
- It shall detect faces using MediaPipe Face Mesh and extract the face region with padding.
- The extracted face region shall be classified by the ResNet-18 model into one of seven emotional states.
- The API response shall include the detected emotion label and its confidence score.

3.2.3 Engagement Scoring

- The system shall estimate head pose (pitch, yaw, roll) using a 3D-2D point correspondence (PnP) solver.
- It shall calculate blink rate from eye-aspect-ratio landmarks.
- A composite engagement score shall be computed by combining emotion weight, head pose deviation, and blink frequency.
- The score shall be updated and returned with each API call.

3.2.4 Session Management

- The system shall persist each session's start time, end time, per-frame emotion logs, and computed engagement scores in MongoDB.

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- Users shall be able to retrieve a history of their past sessions.
- Administrators shall be able to view session data for all users.

3.2.5 Analytics Dashboard

- The frontend shall display a live emotion label and engagement score overlay on the webcam feed.
- It shall render a timeline chart of engagement scores for the current session.
- Emotion distribution shall be shown as a pie/bar chart.
- Session history shall be accessible in a paginated table view.

3.3 Non-Functional Requirements

3.3.1 Performance

- The system shall process a single webcam frame (inference + engagement scoring) within 300 ms under normal load.
- The frontend shall update the emotion overlay at a minimum of 5 frames per second.

3.3.2 Scalability

- The stateless backend design shall allow horizontal scaling by deploying multiple FastAPI instances behind a load balancer.
- MongoDB Atlas shall support automatic scaling of storage.

3.3.3 Security

- All API communication shall occur over HTTPS.
- JWT tokens shall use HS256 signing and include an expiry claim.
- User passwords shall never be stored in plain text.

3.3.4 Usability

- The user interface shall be operable without installation of any additional software or browser extensions.
- The dashboard shall be responsive and usable on screen widths from 768 px (tablet) upwards.

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3.3.5 Reliability

- The system shall gracefully handle frames in which no face is detected, returning a neutral engagement state without crashing.
- The frontend shall display a user-friendly error message if the backend is unreachable.

3.4 System Requirements

3.4.1 Hardware Requirements

Table 3.1: Hardware Requirements

Component	Minimum Specification
-----------	-----------------------

Processor	Intel Core i5 / AMD Ryzen 5 (or equivalent)
-----------	---

RAM	8 GB
-----	------

Webcam	720p USB or built-in camera
--------	-----------------------------

Network	Broadband internet connection (5 Mbps+)
---------	---

Storage (client)	500 MB free disk space (browser cache)
------------------	--

3.4.2 Software Requirements

Table 3.2: Software Requirements

Component	Version / Details
-----------	-------------------

Python	3.10 or higher
--------	----------------

PyTorch	2.x (with torchvision)
---------	------------------------

FastAPI	0.110+
---------	--------

MediaPipe	0.10+
-----------	-------

OpenCV	4.x
--------	-----

MongoDB	Atlas cloud (or local 6.x)
---------	----------------------------

React	18.x
-------	------

Node.js	18 LTS
---------	--------

Web Browser	Chrome 110+ / Firefox 110+ / Edge 110+
-------------	--

3.5 Constraints and Assumptions

- The system assumes adequate and consistent lighting for reliable face detection. Poor lighting may reduce model accuracy.

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- **The ResNet-18 model is trained exclusively on the RAF-DB dataset; performance may vary for demographics underrepresented in that dataset.**
- **Browser webcam access requires the user to grant camera permissions; denial will prevent system operation.**
- **Latency is subject to network conditions between the client and the cloud-hosted backend.**

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CHAPTER 4
SYSTEM DESIGN

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4.1 System Architecture

MindTrace follows a three-tier architecture: Presentation (React + Vite), Application (FastAPI + ResNet-18), and Data (MongoDB Atlas).

Figure 4.1: Architecture Diagram

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4.2 Data Flow Diagram

4.2.1 DFD Level 0

Figure 4.2: DFD Level 0

4.2.2 DFD Level 1

Figure 4.3: DFD Level 1

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4.3 UML Diagrams

4.3.1 Use Case Diagram

Figure 4.4: Use Case Diagram

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4.3.2 State Diagram

Figure 4.5: State Diagram

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4.3.3 Class Diagram

Figure 4.6: Class Diagram

4.4 Database Design

4.4.1 Users Collection

Table 4.1: Users Collection Schema

Field Type Description

_id ObjectId Auto-generated primary key

username String Unique display name

email String Unique email address

hashed_password String bcrypt hash of password

role String student or admin

created_at DateTime Account creation timestamp

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4.4.2 Sessions Collection

Table 4.2: Sessions Collection Schema

Field Type Description

_id ObjectId Auto-generated session ID
user_id ObjectId Reference to Users collection
start_time DateTime Session start timestamp
end_time DateTime Session end timestamp
frames Array Per-frame analysis records
avg_engagement Float Mean engagement score
dominant_emotion String Most frequent emotion

4.5 Module Design

4.5.1 Frontend Modules

- **AuthModule** — Login, registration, JWT storage, route guards.
- **WebcamModule** — Camera access, frame capture, Base64 encoding.
- **AnalysisModule** — Polls /api/analyze and dispatches results.
- **DashboardModule** — Live overlay, timeline chart, pie chart, controls.
- **HistoryModule** — Paginated session history table.
- **AdminModule** — Aggregated reports for all users.

4.5.2 Backend Modules

- **auth_router** — /register and /login endpoints.
- **analysis_router** — /api/analyze and session CRUD.
- **emotion_service** — MediaPipe + ResNet-18 inference pipeline.
- **engagement_service** — PnP head pose and blink-rate (EAR).
- **db_service** — Async MongoDB via Motor driver.
- **security_service** — JWT signing and bcrypt hashing.

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4.6 Algorithm Design

4.6.1 Emotion Recognition Pipeline

- 1. MediaPipe Face Mesh extracts 468 landmarks; bounding box computed with 20% padding.**
- 2. Face crop resized to 224×224 , normalised with ImageNet stats.**
- 3. ResNet-18 produces 7-dim softmax vector; argmax = emotion label.**

4.6.2 Head Pose Estimation

Six canonical 3D landmarks matched to 2D MediaPipe coordinates via cv2.solvePnP. Rodrigues formula converts rotation vector to Euler angles. Distraction flagged when $|\text{yaw}| > 20$ or $|\text{pitch}| > 15$.

4.6.3 Engagement Score

$$E = 0.5 Se + 0.35 Sp + 0.15 Sb \quad [0, 100]$$

where Se=emotion score, Sp=pose score, Sb=blink score.

4.7 Security Design

- JWT Bearer tokens required for all protected endpoints.**
- bcrypt (work factor 12) for password storage.**
- HTTPS enforced via Vercel and Hugging Face TLS.**
- Pydantic models validate all incoming request payloads.**

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5.1 Development Methodology

MindTrace was developed using an Agile-inspired iterative approach. The project was divided into short development cycles (sprints), each producing a testable increment of the system. This allowed the team to validate individual components — such as the emotion model, backend API, and frontend dashboard — independently before full system integration.

The key principles followed were:

- Iterative delivery: Working features were demonstrated at the end of each sprint rather than waiting for complete system development.
- Continuous integration: Code changes were merged frequently and tested to detect integration issues early.
- Modular design: Each system component (model, API, database, frontend) was designed to be independently replaceable.
- Regular review: Weekly team meetings were held to assess progress, resolve blockers, and re-prioritise tasks.

5.2 Team Structure and Roles

The project team consisted of five members, with responsibilities distributed as follows:

Table 5.1: Team Roles and Responsibilities

Member	Role	Primary Responsibilities
Sankalp Gaikwad	Integration & DevOps and Research	Deployment, testing, report writing
Hemant Choudhary	Backend Developer	FastAPI, authentication, API logic
Shraddha Dashpute	Database Manager	MongoDB schema, data persistence
Priyanshu Jadhav	ML Engineer	Model training, RAF-DB preprocessing
Balaji Motkulwar	Frontend Developer	React dashboard, webcam integration

5.3 Project Milestones

The project was executed over approximately six months (September 2025 to February 2026). The key milestones are listed below:

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Table 5.2: Project Milestones

Phase Milestone Target Month

Phase 1 Requirements gathering and literature review September 2025

Phase 2 Dataset preparation and model training October 2025

Phase 3 Backend API development and testing November 2025

Phase 4 Frontend dashboard development December 2025

Phase 5 System integration and end-to-end testing January 2026

Phase 6 Deployment, optimisation, and report submission February 2026

5.4 Task Breakdown

5.4.1 Phase 1 — Requirements and Research

- Study of existing emotion recognition systems and benchmarks.
- Selection of the RAF-DB dataset and ResNet-18 architecture.
- Finalisation of system requirements and technology stack.
- Drafting of SRS document.

5.4.2 Phase 2 — Model Development

- Download and preprocessing of the RAF-DB dataset (15,339 images, 7 emotion classes).
- Implementation of data augmentation pipeline (random flips, rotations, colour jitter).
- Fine-tuning ResNet-18 (pretrained on ImageNet) for 7-class emotion classification.
- Model evaluation and hyperparameter tuning to achieve target accuracy (>85%).

5.4.3 Phase 3 — Backend Development

- Design and implementation of FastAPI REST endpoints.
- Integration of MediaPipe Face Mesh for landmark detection.
- Implementation of head pose estimation and blink detection.
- Engagement score algorithm development and calibration.

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- **JWT authentication and bcrypt password hashing implementation.**
- **MongoDB schema design and Motor-based async database service.**

5.4.4 Phase 4 — Frontend Development

- **Scaffolding of React + Vite project with component architecture.**
- **Webcam capture module with configurable frame-rate control.**
- **Live emotion overlay and engagement score display.**
- **Recharts-based timeline and distribution chart components.**
- **Session history view with pagination.**
- **Admin panel for multi-user engagement monitoring.**

5.4.5 Phase 5 — Integration and Testing

- **End-to-end integration testing of frontend-to-backend communication.**
- **Performance benchmarking (latency, throughput).**
- **Cross-browser compatibility testing (Chrome, Firefox, Edge).**
- **Security testing (JWT validation, input sanitisation).**
- **Bug fixes and performance optimisation.**

5.4.6 Phase 6 — Deployment and Documentation

- **Backend deployment to Hugging Face Spaces via Docker container.**
- **Frontend deployment to Vercel with environment variable configuration.**
- **Final end-to-end system validation in production environment.**
- **Completion of project report and preparation of presentation.**

5.5 Risk Management

Potential risks identified during planning and the mitigation strategies adopted are listed in the table below:

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Table 5.3: Risk Register

Risk Severity Mitigation Strategy

Model accuracy be-

low target

High Explore data augmentation, class bal-
ancing, and alternative architectures

(MobileNetV2)

Backend deploy-

ment failure

Medium Use Docker for reproducibility; main-
tain a local fallback environment

Dataset licensing is-

sues

Medium Use only publicly available, research-
licensed datasets (RAF-DB)

Webcam access de-

nied by browser

Low Prompt user with clear permission
request and fallback error message

Network latency de-

grading UX

Medium Implement client-side frame buffering
and graceful timeout handling

Team member un-

availability

Low Cross-train team members on adja-
cent modules

5.6 Tools and Technologies Used

Table 5.4: Development Tools

Tool / Technology Purpose

Python 3.10 Backend development and ML training

PyTorch 2.x Deep learning model implementation

FastAPI REST API framework

React 18 + Vite Frontend SPA framework

MongoDB Atlas Cloud database

Hugging Face Spaces Backend cloud deployment

Vercel Frontend cloud deployment

Docker Backend containerisation

Git + GitHub Version control and collaboration

VS Code Primary IDE

Postman API testing

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CHAPTER 6
PROJECT IMPLEMENTATION
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6.1 Deep Learning Model Training

6.1.1 Dataset Preparation

The RAF-DB (Real-world Affective Faces Database) dataset was used to train the emotion recognition model. It contains 15,339 facial images annotated with seven basic emotion categories: Angry, Disgust, Fear, Happy, Neutral, Sad, and Surprise. The dataset was split into 12,271 training images and 3,068 test images following the official partition.

Preprocessing steps applied to each image:

- Resize to 224×224 pixels (ResNet-18 input size).
- Random horizontal flip (probability 0.5) for data augmentation.
- Random rotation in the range $[-10, +10]$.
- Colour jitter (brightness, contrast, saturation variation of 0.2).
- Normalisation using ImageNet mean $[0.485, 0.456, 0.406]$ and standard deviation $[0.229, 0.224, 0.225]$.

Class imbalance in the dataset (Disgust and Fear are underrepresented) was addressed using weighted random sampling during training, where the sampling weight of each class is inversely proportional to its frequency.

6.1.2 Model Architecture

ResNet-18 (Residual Network with 18 layers) was selected as the base architecture. It was pretrained on ImageNet and fine-tuned for the 7-class emotion classification task. The key modification was replacing the final fully-connected layer (originally 1000 output units for ImageNet) with a new linear layer of 7 output units:

- Input: $3 \times 224 \times 224$ RGB image tensor
- Feature Extractor: ResNet-18 convolutional backbone (frozen for first 3 epochs, then fully fine-tuned)
- Classifier Head: Linear(512, 7) + Softmax

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6.1.3 Training Configuration

Table 6.1: Model Training Hyperparameters

Hyperparameter Value

Optimiser Adam

Initial learning rate 1×10^3

LR scheduler StepLR (step size = 10, gamma = 0.1)

Batch size 64

Epochs 30

Loss function CrossEntropyLoss

Weight decay 1×10^4

Early stopping patience 5 epochs

Training was performed on a GPU-accelerated environment (NVIDIA CUDA). The best model checkpoint (highest validation accuracy) was saved and exported as a .pth file for deployment.

6.1.4 Model Performance

The trained ResNet-18 model achieved the following results on the RAF-DB test set:

Table 6.2: Model Evaluation Results on RAF-DB Test Set

Emotion Precision Recall

Angry 0.81 0.78

Disgust 0.74 0.70

Fear 0.76 0.73

Happy 0.93 0.95

Neutral 0.88 0.89

Sad 0.82 0.80

Surprise 0.87 0.86

Overall Accuracy 85.7%

6.2 Backend Implementation

6.2.1 Project Structure

The FastAPI backend is organised as follows:

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- **main.py** — FastAPI application entry point; registers routers and CORS middleware.
- **routers/auth.py** — Authentication endpoints (/register, /login).
- **routers/analysis.py** — Inference endpoint (/api/analyze) and session management.
- **services/emotion_service.py** — MediaPipe and ResNet-18 inference logic.
- **services/engagement_service.py** — Head pose and blink-rate computation.
- **services/db_service.py** — Async MongoDB operations.
- **models/schemas.py** — Pydantic request/response models.
- **core/security.py** — JWT and bcrypt utilities.

6.2.2 Emotion Inference Endpoint

The core inference endpoint accepts a Base64-encoded image frame and returns the emotion prediction and engagement metrics:

1. The Base64 payload is decoded to a NumPy array using OpenCV.
2. MediaPipe Face Mesh processes the frame to extract 468 facial landmarks and compute a face bounding box.
3. The face region is cropped, resized to 224×224 , and normalised before being passed to the ResNet-18 model.
4. The model returns a softmax probability vector; the argmax class is selected as the predicted emotion.
5. In parallel, head pose is estimated from 6 key landmarks using cv2.solvePnP, and blink rate is computed from the Eye Aspect Ratio (EAR) of both eyes.
6. The composite engagement score is computed and the full result is serialised as a JSON response.

6.2.3 Authentication Implementation

User registration stores the bcrypt-hashed password (work factor 12) in MongoDB. Login verifies the password hash and issues a signed JWT token with a configurable expiry (default: 60 minutes). All protected API routes use a FastAPI dependency (get_current_user) that decodes and validates the token before passing the user context to the handler.

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6.2.4 CORS and Deployment Configuration

Cross-Origin Resource Sharing (CORS) middleware is configured to allow requests from the Vercel-hosted frontend domain. The application is containerised using Docker with a uvicorn ASGI server and deployed to Hugging Face Spaces via a Dockerfile that installs all Python dependencies and copies the trained model weights.

6.3 Frontend Implementation

6.3.1 Project Structure

The React + Vite frontend is structured using a feature-based layout:

- `src/pages/` — Top-level page components (Login, Register, Dashboard, History, Admin).
- `src/components/` — Reusable UI components (WebcamFeed, EmotionOverlay, EngagementChart, SessionTable).
- `src/services/api.js` — Axios instance with JWT interceptor for attaching the Bearer token to every request.
- `src/store/` — Global state management using React Context for authentication and session state.
- `src/hooks/` — Custom hooks (useWebcam, useAnalysis) for webcam access and API polling.

6.3.2 Webcam Capture Module

The useWebcam hook uses the navigator.mediaDevices.getUserMedia Web API to access the camera stream. Frames are captured at a configurable interval (200 ms default) by drawing the video stream onto a hidden `<canvas>` element and calling `canvas.toDataURL('image/jpeg', 0.7)` to produce a compressed Base64 string for transmission.

6.3.3 Live Dashboard

The main dashboard renders:

- A live webcam feed with the detected emotion label and confidence overlaid as a transparent badge.
- An engagement score gauge updated in real time.

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- A Recharts LineChart plotting engagement score over the session timeline.
 - A PieChart showing the distribution of detected emotions for the current session.
- Session start/stop controls trigger API calls to create and finalise session records in MongoDB.

6.4 Database Integration

The backend uses Motor, the **asynchronous Python driver for MongoDB**, to perform non-blocking database operations. A connection pool is initialised at application startup and shared across all request handlers via FastAPI's dependency injection system. Key operations include:

- User creation: Insert a new document into the users collection on registration.
- Session creation: Insert a new session document on session start.
- Frame append: Use MongoDB's \$push operator to append each frame analysis record to the session's frames array atomically.
- Session finalisation: Update end_time, avg_engagement, and dominant_emotion on session end.
- History retrieval: Query sessions by user_id with pagination using skip and limit.

6.5 Deployment

6.5.1 Backend Deployment — Hugging Face Spaces

The FastAPI backend is deployed as a Docker Space on Hugging Face. The Dockerfile performs the following steps:

1. Uses python:3.10-slim as the base image.
2. Installs system dependencies (libGL for OpenCV).
3. Copies and installs Python packages from requirements.txt.
4. Copies the application source and pre-trained model weights.
5. Exposes port 7860 and starts the Uvicorn server.

Environment variables (MongoDB URI, JWT secret key) are configured through the Hugging Face Spaces secrets interface to avoid hardcoding sensitive credentials.

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6.5.2 Frontend Deployment — Vercel

The React application is deployed on Vercel by connecting the GitHub repository. Vercel automatically builds the Vite project on each push to the main branch using the command `npm run build`. The backend API URL is configured as a Vercel environment variable (`VITE_API_URL`) and injected at build time.

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CHAPTER 7

TESTING

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7.1 Testing Strategy

The testing approach for MindTrace followed a bottom-up strategy:

- 1. Unit Testing — Individual functions and modules were tested in isolation to verify correct behaviour.
- 2. Integration Testing — Interactions between modules (e.g., frontend-to-API, API-to-database) were verified.
- 3. System Testing — The complete end-to-end system was tested against the requirements defined in the SRS.
- 4. Performance Testing — Latency and throughput were measured under simulated load.
- 5. User Acceptance Testing (UAT) — The system was evaluated by end users to validate usability and correctness.

7.2 Unit Testing

7.2.1 Emotion Service

Table 7.1: Unit Test Cases — Emotion Service

Test Case	Input	Expected Output	Result
Valid face frame inference	224×224 frontal face image	Emotion label + confidence	Pass
No face in frame	Blank/background image	null emotion, score = 50	Pass
Multiple faces in frame	Image with 2 faces	Primary face detected	Pass
Corrupt / malformed image			
Invalid Base64 string	HTTP 400 error response		Pass

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7.2.2 Engagement Service

Table 7.2: Unit Test Cases — Engagement Service

Test Case	Input	Expected Output	Result
Front-facing head pose	Yaw = 0°, Pitch = 0°	High pose score (>80)	Pass
Side-facing head pose	Yaw = 30°	Low pose score (<40)	Pass
Normal blink rate	EAR oscillation pattern	Normal blink score	Pass
Engagement score range	Any emotion + pose	Score in [0, 100]	Pass

7.2.3 Authentication Service

Table 7.3: Unit Test Cases — Authentication Service

Test Case	Input	Expected Output	Result
Valid user registration	New username, email, password	User document created	Pass
Duplicate email registration	Existing email	HTTP 409 Conflict	Pass
Valid login	Correct credentials	JWT token returned	Pass
Invalid password login	Wrong password	HTTP 401 Unauthorized	Pass
Expired token access	Expired JWT	HTTP 401 Unauthorized	Pass

7.3 Integration Testing

7.3.1 Frontend to Backend Integration

Integration tests verified that the React frontend correctly communicates with the FastAPI backend under real network conditions.

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Table 7.4: Integration Test Cases — Frontend to Backend

Scenario Expected Behaviour Result

Login from frontend JWT stored in localStorage Pass

Webcam frame sent to

/api/analyze

Emotion + score displayed in

UI

Pass

Session start API call Session document created in

MongoDB

Pass

Session end API call Session summary updated in

MongoDB

Pass

Unauthenticated API call Redirect to login page Pass

Backend unreachable Error toast displayed to user Pass

7.3.2 Backend to Database Integration

Table 7.5: Integration Test Cases — Backend to MongoDB

Scenario Expected Behaviour Result

User document insert Document found in users collection

Pass

Session document insert Document found in sessions collection

Pass

Frame \$push to session Frame appended to frames array

Pass

Session history query by user

ID

Correct sessions returned Pass

Pagination (skip / limit) Correct subset of sessions returned

Pass

7.4 System Testing

System testing validated end-to-end user flows against the functional requirements specified in the SRS.

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Table 7.6: System Test Cases — End-to-End Flows

Test Scenario	Steps	Expected Result	Result
New user registration and login	Register	Login	
Dashboard renders correctly	Dashboard		
Live emotion monitoring	Start session	face web-cam	
Emotion label updates live			
Engagement score tracking	Perform focus/distraction actions		
Score reflects state changes			
Session history view	End session	view History	
Past session visible in table			
Admin multi-user view	Login as admin	Admin panel	
All user sessions visible			
Role-based access control	Student accesses admin route		
Access denied (403)			Pass

7.5 Performance Testing

Performance tests were conducted to measure system responsiveness under realistic usage conditions.

7.5.1 API Latency

Latency was measured for the /api/analyze endpoint by sending 100 sequential requests with a standard 224×224 face image:

Table 7.7: API Latency Measurements

Metric	Value
Mean response time	187 ms

Median response time 172 ms

95th percentile latency 264 ms

Maximum response time 312 ms

Throughput 5.3 requests/second

All measurements were within the 300 ms performance requirement defined in the SRS. The system supports the target minimum update rate of 5 frames per second for the live dashboard.

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7.5.2 Cross-Browser Compatibility

The frontend was tested across major browsers to ensure consistent behaviour:

Table 7.8: Cross-Browser Compatibility Results

Browser Version Result

Google Chrome 122 Pass

Mozilla Firefox 123 Pass

Microsoft Edge 122 Pass

Safari (macOS) 17 Pass

7.6 User Acceptance Testing

User Acceptance Testing (UAT) was conducted with a small group of five student volunteers who interacted with the deployed MindTrace system for a 15-minute session each. Participants were asked to rate the system on a scale of 1 (poor) to 5 (excellent) across four criteria:

Table 7.9: User Acceptance Testing Results

Criterion Average Score (out of 5)

Ease of use and interface clarity 4.4

Accuracy of detected emotions 4.0

Responsiveness of the live dashboard 4.2

Overall system usefulness 4.3

Overall Average 4.2

Key feedback collected from participants:

- The live emotion overlay was intuitive and easy to understand.
- Emotion detection accuracy was perceived as reliable under good lighting conditions.
- Some users noted occasional misclassification between Neutral and Sad expressions.
- The session history view was considered a useful feature for self-review.

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7.7 Defects and Resolutions

Table 7.10: Defects Identified and Resolved During Testing

Defect Resolution Status

No face detected —

system crashes

Added null check be-

fore inference call

Resolved

JWT token not re-

freshed on expiry

Added 401 interceptor

in Axios

Resolved

Engagement score ex-

ceeds 100 in edge case

Added clamp(0, 100)

to score formula

Resolved

MongoDB connection

timeout under load

Configured connection

pooling (max=10)

Resolved

CORS error on cross-

origin frame request

Whitelisted Vercel do-

main in CORS config

Resolved

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CHAPTER 8

RESULTS

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8.1 Emotion Recognition Results

8.1.1 Model Training Convergence

The ResNet-18 model was trained for 30 epochs on the RAF-DB dataset. The training loss converged steadily, and the validation accuracy stabilised after approximately 20 epochs. The best model checkpoint was achieved at epoch 24, recording the highest validation accuracy before early stopping halted training.

Table 8.1: Training and Validation Metrics at Key Epochs

Epoch	Training Loss	Val. Accuracy (%)	Val. Loss
5	1.42	71.3	1.18
10	0.98	78.6	0.89
15	0.74	82.1	0.74
20	0.58	84.3	0.68
24 (best)	0.48	85.7	0.63
30	0.43	85.4	0.65

8.1.2 Per-Class Classification Performance

The confusion matrix analysis revealed that the model performs best on Happy (93% precision) and Neutral (88% precision), which are the most frequently occurring classes in the RAF-DB dataset. Disgust and Fear showed relatively lower performance due to class imbalance, despite the use of weighted sampling.

Table 8.2: Per-Class F1 Scores on RAF-DB Test Set

Emotion	Precision	Recall	F1-Score
Angry	0.81	0.78	0.79
Disgust	0.74	0.70	0.72
Fear	0.76	0.73	0.74
Happy	0.93	0.95	0.94
Neutral	0.88	0.89	0.88
Sad	0.82	0.80	0.81
Surprise	0.87	0.86	0.86
Weighted Avg.	0.86	0.857	0.857

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8.1.3 Comparison with Existing Approaches

The performance of the MindTrace ResNet-18 model is compared against representative methods from the literature reviewed in Chapter 2:

Table 8.3: Comparison of Emotion Recognition Accuracy on RAF-DB

Method Architecture Accuracy (%)

Li et al. (2017) DLP-CNN 84.13

Wang et al. (2020) SCN 87.03

Zhao et al. (2021) TransFER (Transformer) 90.91

MindTrace (Ours) ResNet-18 fine-tuned 85.70

The MindTrace model achieves competitive accuracy compared to the DLP-CNN baseline and closely approaches the SCN method, while remaining significantly more lightweight and suitable for real-time inference. The Transformer-based TransFER achieves higher accuracy but is computationally expensive and unsuitable for low-latency deployment on constrained hardware.

8.2 Engagement Scoring Results

8.2.1 Score Behaviour Under Controlled Conditions

The engagement scoring algorithm was validated by having participants perform predefined actions while the system was running. The observed engagement scores matched expected behaviour across all tested conditions:

Table 8.4: Engagement Score Observations Under Controlled Conditions

Participant Action Observed Score Range

Looking directly at screen (neutral expression) 78 – 85

Smiling / happy expression, facing camera 88 – 96

Looking away (yaw > 25°) 30 – 45

Eyes closed / frequent blinking 40 – 55

Visible frustration / angry expression 55 – 65

Combined distraction (look away + tired) 20 – 35

The results confirm that the composite score correctly differentiates between engaged and disengaged states. The highest scores correspond to direct eye contact with a

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positive emotional state, while combined distraction cues (poor head pose and high blink rate) produce the lowest scores, as expected.

8.2.2 Real-Time Session Timeline

A typical 10-minute session produces a continuous timeline of engagement scores.

Analysis of five recorded sessions showed:

- Average session engagement: 72.4 out of 100
- Most frequent emotion detected: Neutral (43% of frames)
- Second most frequent: Happy (28% of frames)
- Distraction events (score < 40): 6.2% of frames on average

8.3 System Performance Results

8.3.1 API Response Latency

End-to-end latency was measured from the moment the frontend sends a frame to when the response is rendered on the dashboard:

Table 8.5: End-to-End Latency Breakdown

Stage Average Time (ms)

Frame capture and Base64 encoding 12

Network transmission (client to server) 38

Face detection (MediaPipe) 28

Emotion model inference (ResNet-18) 95

Engagement computation 14

Network transmission (server to client) 31

Total End-to-End Latency 218 ms

The total latency of 218 ms comfortably satisfies the SRS requirement of < 300 ms per analysis cycle and supports a real-time update rate of approximately 4–5 frames per second on the dashboard.

8.3.2 System Scalability

The stateless backend design was stress-tested by simulating concurrent users using the Locust load-testing framework:

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Table 8.6: Scalability Test Results

Concurrent Users	Mean Response Time (ms)	Error Rate (%)
1	187	0.0
5	204	0.0
10	243	0.0
20	298	0.8
50	512	4.2

The system handles up to 20 concurrent users within acceptable latency bounds. Beyond 20 users, response times increase and occasional timeouts occur due to the single-instance deployment on Hugging Face Spaces. Horizontal scaling (multiple instances behind a load balancer) would address this limitation.

8.4 System User Interface

This section showcases the final user interface of the MindTrace application, demonstrating the user journey from authentication to real-time analytics.

8.4.1 Authentication and Onboarding

The system provides a modern, responsive interface for user onboarding. Figure 8.1 shows the landing page and the secure login/signup modules.

(a) MindTrace Landing Page (b) Secure Login Interface

Figure 8.1: System Onboarding and Authentication

8.4.2 Live Monitoring Dashboard

The core functionality of MindTrace is demonstrated through the live dashboard. Figure 8.2 captures the system performing real-time emotion inference and focus tracking via the student’s webcam.

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Figure 8.2: Real-Time Monitoring with Emotion Overlay

8.4.3 Session Analytics and History

After each session, the system provides detailed behavioural insights. Figure 8.3 shows the post-session dashboard featuring emotion distribution and engagement trends.

Figure 8.3: Post-Session Behavioural Analytics Dashboard

8.5 Cloud Deployment and Database

MindTrace is deployed as a distributed cloud application to ensure global accessibility and reliable data persistence.

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(a) Frontend Hosting on Vercel (b) Backend API on Hugging Face

Figure 8.4: Cloud Deployment Infrastructure

The session data is persisted in a MongoDB Atlas cluster, as shown in Figure 8.5, which allows for real-time data exploration and long-term storage of student engagement records.

Figure 8.5: MongoDB Atlas Data Explorer (Session Records)

8.6 Discussion

8.6.1 Achievement of Objectives

All primary objectives defined in Chapter 1 were successfully achieved:

- **Emotion classification:** Seven-class emotion detection with 85.7% accuracy was implemented and deployed in real time.
- **Head pose and blink detection:** Accurately flags distraction events with a low false-positive rate.

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- **Engagement scoring:** The composite score provides meaningful and interpretable feedback that correlates with observed participant behaviour.
- **Data persistence:** Session records are reliably stored and retrievable from MongoDB Atlas.
- **Dashboard:** A fully functional React dashboard with live overlays, charts, and admin panel was delivered.
- **Security:** JWT authentication and bcrypt password hashing were implemented and validated.
- **Cloud deployment:** The system is live and accessible via public URLs on Hugging Face Spaces and Vercel.

8.6.2 Limitations

- The model exhibits reduced accuracy in low-lighting conditions, as the RAF-DB training images are primarily well-lit.
- Disgust and Fear classes have lower F1 scores due to underrepresentation in the dataset.
- The engagement score weights (w_e , w_p , w_b) were set empirically and may not generalise equally across all demographic groups.
- The single-instance deployment limits concurrent user capacity to approximately 20 users.

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CHAPTER 9
CONCLUSION
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9.1 Conclusion

This project successfully designed, implemented, and deployed MindTrace — a real-time emotion recognition and engagement tracking system that operates entirely through a standard webcam without requiring any specialised hardware. The system demonstrates that modern deep learning techniques can be effectively applied to affective computing in educational and professional environments.

The key outcomes of the project are summarised as follows:

- A ResNet-18 model fine-tuned on the RAF-DB dataset achieved 85.7% accuracy in classifying seven emotional states, establishing a strong foundation for real-time affective feedback.
- MediaPipe Face Mesh enabled robust, landmark-based face detection that operates reliably at real-time frame rates without a GPU on the client side.
- The composite engagement scoring algorithm — combining emotional state, head pose estimation, and blink-rate analysis — produced scores that accurately reflected participant behaviour under controlled and real-world conditions.
- A stateless FastAPI backend deployed on Hugging Face Spaces demonstrated cloud-native design principles, supporting concurrent access and enabling hardware-independent operation.
- A React-based analytics dashboard provided an intuitive interface for live monitoring, session history, and multi-user administration, validated through user acceptance testing with an overall score of 4.2 out of 5.
- The system meets all functional and non-functional requirements defined in the SRS, including the 300 ms latency target, HTTPS security, role-based access control, and cross-browser compatibility.

MindTrace bridges the gap between affective computing research and practical, deployable applications. By providing objective, continuous engagement feedback, it offers educators, corporate trainers, and researchers a powerful tool for understanding how people interact with digital learning environments — without the need for wearable sensors, invasive monitoring, or manual observation.

9.2 Future Scope

While MindTrace achieves its defined objectives, several directions remain open for further enhancement:

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- 1. Multi-person Monitoring:** Extending the system to track multiple faces simultaneously within a single frame would enable classroom-level engagement monitoring, where a single camera covers all participants.
 - 2. Larger and More Diverse Datasets:** Training on a more diverse dataset — such as AffectNet (450,000+ images) or a combination of RAF-DB and CK+ — could improve model robustness across different ethnicities, ages, and lighting conditions.
 - 3. Temporal Emotion Modelling:** Incorporating temporal context using LSTM or Transformer-based sequence models could capture dynamic emotional transitions more accurately than frame-level classification alone.
 - 4. Voice and Physiological Fusion:** Integrating audio sentiment analysis (tone of voice) or physiological signals (heart rate via remote photoplethysmography) could produce a more holistic and accurate engagement metric.
 - 5. On-Device Inference:** Optimising the ResNet-18 model using quantisation or knowledge distillation and deploying it as a TensorFlow.js model in the browser would eliminate network latency entirely and improve privacy by keeping all processing client-side.
 - 6. Personalised Engagement Baselines:** Rather than using fixed score weights, the system could learn individualised engagement baselines over time using online learning, adapting to each user's natural expression and attention patterns.
 - 7. LMS Integration:** Integrating MindTrace with popular Learning Management Systems such as Moodle or Google Classroom via APIs would allow engagement data to be linked directly to course content and assessment performance.
 - 8. Scalable Deployment:** Deploying the backend on a container orchestration platform such as Kubernetes with auto-scaling would remove the current single-instance concurrency limitation and support large-scale institutional deployments.
- In summary, MindTrace lays a solid foundation for intelligent, non-invasive engagement monitoring. Its modular, cloud-native architecture ensures that each of the above enhancements can be adopted incrementally without requiring a complete redesign of the system.

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CHAPTER 10
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ANNEXURE

A. API Endpoint Reference

Table 10.1: MindTrace REST API Endpoints

Method	Endpoint	Auth	Description
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POST	/register	No	Register a new user account
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POST	/login	No	Login and receive JWT token
------	--------	----	-----------------------------

POST	/api/analyze	Yes	Submit webcam frame for analysis
------	--------------	-----	----------------------------------

POST	/api/session/start	Yes	Create a new session record
------	--------------------	-----	-----------------------------

POST	/api/session/end	Yes	Finalise and save current session
------	------------------	-----	-----------------------------------

GET	/api/sessions	Yes	Retrieve current user's sessions
-----	---------------	-----	----------------------------------

GET	/api/admin/users	Admin	Retrieve all user records
-----	------------------	-------	---------------------------

GET	/api/admin/sessions	Admin	Retrieve all sessions (all users)
-----	---------------------	-------	-----------------------------------

B. Sample API Request and Response

Request — POST /api/analyze:

```
{  
  "frame": "<base64-encoded JPEG string>",  
  "session_id": "6614f3a2c3e4b12a9d7f0011"  
}
```

Response:

```
{  
  "emotion": "Happy",  
  "confidence": 0.91,  
}
```

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```
"engagement_score": 87,  
"head_pose": {  
  "pitch": -2.3,  
  "yaw": 4.1,  
  "roll": 1.8  
},  
"blink_rate": 14  
}
```

C. Technology Stack Summary

Table 10.2: Complete Technology Stack

Layer	Technology	Version
ML Framework	PyTorch + torchvision	2.x
Model Architecture	ResNet-18 (fine-tuned)	—
Face Detection	MediaPipe Face Mesh	0.10+
Computer Vision	OpenCV	4.x
API Framework	FastAPI + Uvicorn	0.110+
Database	MongoDB Atlas + Motor	6.x
Authentication	PyJWT + bcrypt	—
Frontend Framework	React + Vite	18.x / 5.x
Charting Library	Recharts	2.x
HTTP Client	Axios	1.x
Backend Deployment	Hugging Face Spaces (Docker)	—
Frontend Deployment	Vercel	—
Version Control	Git + GitHub	—

D. Emotion Score Weights

The engagement score is computed using the following empirically determined weights applied to the three signal components:

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Table 10.3: Engagement Score Component Weights

Component Weight Rationale

Emotion score (Se) 0.50 Primary indicator of cognitive state

Head pose score (Sp) 0.35 Strong proxy for visual attention

Blink rate score (Sb) 0.15 Supplementary fatigue indicator

Total 1.00 —

